

A Microscopic Model for the Dependence of Vacuum Energy on Matter Density: Toward a Quantum-Field-Theoretic Interpretation

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Abstract

The cosmological constant problem arises from the identification of the quantum vacuum energy density ρ_{vac} with the cosmological constant Λ , producing a discrepancy of 55–120 orders of magnitude between theoretical estimates and observational bounds. A growing body of work suggests that this identification is not a theorem of established physics but an additional hypothesis. Building on these developments, this paper proposes and analyzes a microscopic mechanism by which the effective vacuum energy density becomes a function of the local matter density, $\rho_{\text{vac}} = \rho_{\text{vac}}(\rho_{\text{m}})$, arising naturally from quantum fields propagating in curved spacetime in the presence of matter sources. We develop the framework from the one-loop effective action in curved spacetime, employing the Schwinger–DeWitt heat-kernel expansion to show that matter-induced curvature modifies the spectrum of vacuum fluctuations and thereby the renormalized vacuum energy density. The leading correction takes the form $\Delta\rho_{\text{vac}} \approx -\alpha \rho_{\text{m}}$, where α is a dimensionless coupling coefficient determined by loop factors and the field content. We prove that such a linear response is the generic leading-order behavior under the assumptions of perturbative QFT on mildly curved backgrounds, estimate $\alpha \sim 10^{-2}$ from standard one-loop factors, and discuss the physical interpretation in terms of matter-induced vacuum polarization. We show that this mechanism is consistent with the running vacuum model of Solà Peracaula and collaborators, with the separation thesis developed in the companion Paper 1a, and we identify a structural analogy with Brown’s Quantum-Kinetic Dark Energy (QKDE) framework. Falsifiable predictions

and observational implications for late-time cosmology are identified. The model is offered as a first step toward bridging phenomenological vacuum-energy models and the microscopic physics of quantum fields in curved spacetime.

Keywords: vacuum energy, matter density, cosmological constant problem, effective action, curved spacetime, vacuum polarization, running vacuum model, Schwinger–DeWitt expansion, dark energy, effective field theory

Contents

1	Introduction	5
2	Logical Structure of the Argument	6
3	Background and Related Work	7
3.1	The cosmological constant problem	7
3.2	Environment-dependence of vacuum energy	7
3.3	Running vacuum models	7
3.4	Volovik’s condensed-matter analogy	8
3.5	Brown’s QKDE framework	8
4	Conceptual Framework	9
5	Mathematical Development I: The One-Loop Effective Action and Heat-Kernel Formalism	10
5.1	The effective action in curved spacetime	10
5.2	The Schwinger–DeWitt heat-kernel expansion	11
5.3	Renormalized effective Lagrangian	12
6	Mathematical Development II: Derivation of the Matter-Dependent Vacuum Correction	13
6.1	From curvature-dependence to matter-dependence	13
6.2	Generalization to the Standard Model field content	14
6.3	Estimation of the coupling coefficient α	15
7	Mathematical Development III: Connection to Running Vacuum and Brown’s QKDE Frameworks	18
7.1	Structural analogy with the running vacuum model	18
7.2	Structural analogy with Brown’s QKDE framework	19
7.3	Unified interpretation (present author’s synthesis)	20
8	Empirical and Data-Grounded Illustrations	22
8.1	Modified Friedmann equation	22
8.2	Breaking the degeneracy: perturbation theory	22
8.3	Observational constraints	23
9	Discussion	24
10	Limitations	26

11 Future Directions	27
12 Position Within the Research Program	28
13 Recommended Journal for Publication	29
Acknowledgments	29
References	29
Appendices	33
A Heat-Kernel Coefficients for Higher-Spin Fields	33
B On the Vacuum Polarization Channel: Derivation Status and Open Questions	33
Peer Reviews and Revision History	35

1 Introduction

The cosmological constant problem—the observation that quantum field theory (QFT) estimates of the vacuum energy density exceed the observationally inferred value by 55–120 orders of magnitude [3, 4, 5]—remains one of the deepest unresolved questions in theoretical physics. The standard formulation of the problem rests on the identification, due to Zel’dovich [1, 2], of the quantum vacuum energy density ρ_{vac} with the cosmological constant Λ through the relation $\Lambda = 8\pi G \rho_{\text{vac}}$.

In the companion Paper 1a [28], the present author reviewed five independent lines of evidence—geometric (unimodular gravity), thermodynamic (Volovik’s condensed-matter analogy), field-theoretic (running vacuum models), effective-field-theoretic (Brown’s QKDE framework [26]), and foundational (Rugh–Zinkernagel critique)—establishing that this identification is a physical hypothesis rather than a mathematical theorem. The *separation thesis* holds that Λ and ρ_{vac} are logically and physically distinct quantities, and that the apparent 120-order discrepancy may be an artifact of conflating them.

A separate but complementary phenomenological approach has been developed in Paper 2 of the present research program [29], which proposed that the effective vacuum energy may depend on the local matter density: in regions of higher matter density, the vacuum contribution to the gravitational field is partially suppressed. Such models can account for a range of cosmological observations without invoking exotic particles or extreme fine-tuning. However, the principal weakness of any phenomenological model is the absence of a microscopic derivation.

The present paper addresses this gap. We develop a quantum-field-theoretic framework in which the effective vacuum energy density becomes a function of the local matter density,

$$\rho_{\text{vac}} = \rho_{\text{vac}}(\rho_{\text{m}}), \tag{1}$$

arising naturally from quantum fields propagating in curved spacetime in the presence of matter. The key insight is that matter modifies the local spacetime curvature via Einstein’s equations, which in turn modifies the spectrum of vacuum fluctuations and thereby the renormalized vacuum energy density. This is not a new physical mechanism but a consequence of well-known features of QFT in curved spacetime: the effective action depends on curvature invariants, and curvature is sourced by matter.

Our goal is not to derive a final, exact formula—that would require a complete theory of quantum gravity—but to demonstrate that such a dependence can emerge within the standard framework of QFT in curved spacetime through well-known mechanisms such as effective-action renormalization and matter-induced vacuum polarization, and to estimate the magnitude of the effect.

The paper is organized as follows. Section 2 presents the logical structure of the argument.

Section 3 reviews the relevant background. Section 4 develops the conceptual framework. Sections 5–7 contain the mathematical development in three stages: the one-loop effective action and heat-kernel formalism, the derivation of the matter-dependent correction, and the connection to the running vacuum and Brown’s QKDE frameworks. Section 8 provides empirical illustrations. Sections 9–11 contain the discussion, limitations, and future directions.

2 Logical Structure of the Argument

The overall reasoning proceeds through the following sequence of steps.

Step 1 (Environment-dependence of the vacuum). We recall that the quantum vacuum is not a rigid universal state but depends on boundary conditions, curvature, and the matter content of the environment. Well-known examples—the Casimir effect, the Unruh effect, Hawking radiation—establish this principle within standard QFT.

Step 2 (One-loop effective action in curved spacetime). The vacuum energy density is computed from the one-loop effective action $\Gamma[g]$, which, after renormalization, contains curvature-dependent contributions. The Schwinger–DeWitt heat-kernel expansion provides a systematic tool for organizing these contributions.

Step 3 (Matter as a source of curvature). Through Einstein’s field equations, the presence of matter with density ρ_m determines the local curvature. The Ricci scalar in a matter-dominated region of characteristic density ρ_m satisfies $R \sim 8\pi G \rho_m$.

Step 4 (Matter-induced vacuum correction). Combining Steps 2 and 3, the renormalized vacuum energy density acquires a correction that depends on the matter density. At leading order, this correction is linear: $\Delta\rho_{\text{vac}} \approx -\alpha \rho_m$, where α is a dimensionless coefficient determined by loop factors and the field content.

Step 5 (Estimation of the coupling). Standard one-loop factors from vacuum polarization diagrams yield $\alpha \sim g^2/(16\pi^2) \sim 10^{-2}$ for couplings of order unity, consistent with known perturbative magnitudes and not requiring fine-tuning.

Step 6 (Connection to established frameworks). The matter-dependent vacuum energy is shown to be structurally consistent with the running vacuum model (RVM) of Solà Peracaula and collaborators [18, 19, 21] and with the separation thesis of Paper 1a [28]. In addition, the present author identifies a structural analogy with Brown’s QKDE framework [26], in which a curvature-motivated kinetic normalization $K = 1 + \alpha R/M^2$ produces H^2 -dependent corrections to the dark-energy sector. This analogy is the present author’s interpretive observation; Brown’s original motivation concerns the dynamics of the kinetic sector [26, 27].

Step 7 (Observational implications). The model yields specific, testable predictions for the suppression of vacuum energy in overdense regions and for the behavior of the effective dark-energy equation of state.

Synthesis. The dependence of vacuum energy on matter density is not an ad hoc phenomenological assumption but a natural consequence of perturbative QFT in curved spacetime. The magnitude of the effect is set by standard loop factors, not by Planck-scale physics.

3 Background and Related Work

3.1 The cosmological constant problem

The semiclassical Einstein equation reads

$$G_{\mu\nu} + \Lambda_{\text{bare}} g_{\mu\nu} = 8\pi G \left(\langle T_{\mu\nu} \rangle_{\text{ren}} + T_{\mu\nu}^{(\text{matter})} \right), \quad (2)$$

where $\langle T_{\mu\nu} \rangle_{\text{ren}}$ is the renormalized vacuum expectation value. The effective cosmological constant is $\Lambda_{\text{eff}} = \Lambda_{\text{bare}} + 8\pi G \rho_{\text{vac}}$. The problem is that ρ_{vac} , computed from field-theoretic zero-point sums with a Planck-scale cutoff, vastly exceeds $\Lambda_{\text{eff}}/(8\pi G)$. Specifically, for the Standard Model field content, the one-loop estimate gives $\rho_{\text{vac}} \sim M_{\text{Pl}}^4/(16\pi^2)$, while observation requires $\Lambda_{\text{eff}} \sim 10^{-122} M_{\text{Pl}}^2$ [3, 4].

3.2 Environment-dependence of vacuum energy

The idea that the vacuum state depends on the physical environment is well established within QFT:

The *Casimir effect* [8] demonstrates that modifying the allowed mode spectrum of a quantum field—by introducing conducting plates—changes the vacuum energy density. The measured force between the plates is a direct consequence of this modification [9].

The *Unruh effect* [10] shows that observers in uniformly accelerated frames perceive the Minkowski vacuum as a thermal state with temperature $T = a/(2\pi)$ (in natural units), where a is the proper acceleration.

Hawking radiation [11] demonstrates that the vacuum state near a black hole horizon produces a thermal flux of particles, further illustrating the observer- and geometry-dependence of the vacuum.

These examples establish that the vacuum energy is not a rigid constant but an emergent quantity that depends on boundary conditions and the geometry in which the quantum fields exist.

3.3 Running vacuum models

Moreno-Pulido and Solà Peracaula [19, 20] have computed the renormalized vacuum energy density for quantum fields in FLRW spacetime using adiabatic regularization. Their central

result is that ρ_{vac} evolves with the Hubble rate:

$$\rho_{\text{vac}}(H) = \rho_{\text{vac}}^{(0)} + \frac{3\nu}{8\pi} M_{\text{Pl}}^2 H^2 + \mathcal{O}(H^4), \quad (3)$$

where ν is a small dimensionless coefficient ($|\nu| \sim 10^{-3}$ for the Standard Model). Crucially, no quartic mass contributions $\sim m^4$ appear after proper renormalization [19]. This dynamical vacuum energy density is inconsistent with a strictly constant Λ and provides field-theoretic support for the separation thesis.

3.4 Volovik’s condensed-matter analogy

Volovik [24, 25] demonstrated that in superfluid ^4He at zero temperature, the ground-state energy is determined by thermodynamic equilibrium conditions (the Gibbs–Duhem relation) rather than by the zero-point energies of the low-energy collective modes. The microscopic zero-point contributions cancel exactly against trans-Planckian (atomic-scale) contributions, not by fine-tuning but by thermodynamic stability. By analogy, the cosmological Λ should be determined by macroscopic parameters of the universe— H , G , and the matter content—rather than by the zero-point energy of quantum fields.

3.5 Brown’s QKDE framework

Brown [26] has constructed a Quantum-Kinetic Dark Energy (QKDE) framework in which the effective action is

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} M_{\text{Pl}}^2 R + K(\chi) X - V(\phi) \right], \quad (4)$$

where $X = -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi$ and $K(\chi) > 0$ is a time-dependent kinetic normalization. In the curvature-motivated specification, $K = 1 + \alpha R/M^2$, where α is a dimensionless Wilson coefficient [26]. This construction realizes the separation of Λ from ρ_{vac} at the level of an action principle: the scalar sector carries all dark-energy phenomenology while the Einstein–Hilbert metric sector remains unmodified.

Counterarguments and Responses

Objection: These background frameworks—running vacuum, Brown’s QKDE—describe the cosmological evolution of vacuum energy on homogeneous backgrounds. How does a *local* dependence on matter density connect to them?

Response: The running vacuum model describes $\rho_{\text{vac}}(H)$, where H characterizes the global expansion rate. But in an inhomogeneous universe, the local curvature differs from the background curvature: overdense regions have higher Ricci scalar than the cosmic mean. If

ρ_{vac} depends on curvature (as the RVM demonstrates, and as Brown’s QKDE framework [26] realizes at the action level), then—as the present author observes—it also depends on the local matter density through the Einstein constraint $R \sim 8\pi G \rho_{\text{m}}$. It should be emphasized that Brown’s original motivation for the curvature-motivated kinetic normalization concerns the dynamics of the kinetic sector (wave-function renormalization of a light scalar by heavy fields), not vacuum energy running or local matter-density dependence directly [26, 27]. The connection to local matter density is the present author’s interpretive extension, developed in this paper.

Section Result and Implications

The cosmological constant problem, the environment-dependence of vacuum energy, the running vacuum framework, and Brown’s QKDE construction all provide complementary motivations for investigating a matter-dependent vacuum energy density. The present paper develops the microscopic QFT mechanism underlying such dependence.

4 Conceptual Framework

We now formulate the conceptual structure that organizes the mathematical development.

Definition 4.1 (Bare vacuum energy density). The bare vacuum energy density ρ_{vac}^0 is the renormalized zero-point energy density of quantum fields computed in a background spacetime with no matter sources, using a well-defined regularization and renormalization prescription.

Definition 4.2 (Matter-induced vacuum correction). The matter-induced vacuum correction $\Delta\rho_{\text{vac}}(\rho_{\text{m}})$ is the change in the renormalized vacuum energy density caused by the presence of a matter distribution with energy density ρ_{m} , computed from the one-loop effective action evaluated on the matter-sourced spacetime geometry.

Definition 4.3 (Effective vacuum energy density). The effective vacuum energy density in the presence of matter is

$$\rho_{\text{vac}}(\rho_{\text{m}}) = \rho_{\text{vac}}^0 + \Delta\rho_{\text{vac}}(\rho_{\text{m}}). \quad (5)$$

Remark 4.4 (Central physical hypothesis). The core claim of this paper is that $\Delta\rho_{\text{vac}}(\rho_{\text{m}})$ is nonzero and, at leading order in a perturbative expansion, takes the form

$$\Delta\rho_{\text{vac}} \approx -\alpha \rho_{\text{m}}, \quad (6)$$

where α is a small dimensionless coefficient whose sign and magnitude are determined by the field content and the specific physical channel (see Section 6). We shall establish that the

rigorously derived channel (curvature-dependent effective action) yields $|\alpha| \sim 10^{-34}$ (negligible), while a conjectured but physically motivated channel (direct vacuum polarization) suggests $|\alpha| \sim 10^{-2}$. The sign of α is not uniquely determined by the present analysis: $\alpha > 0$ corresponds to vacuum suppression by matter, while $\alpha < 0$ corresponds to vacuum enhancement. Both possibilities are explored.

The remainder of the paper is devoted to investigating this claim within the framework of one-loop QFT in curved spacetime.

Counterarguments and Responses

Objection: A linear dependence $\Delta\rho_{\text{vac}} \propto -\rho_{\text{m}}$ would modify the Friedmann equations and could conflict with precision cosmological observations.

Response: For the rigorously derived channel (curvature-dependent effective action), $|\alpha| \sim 10^{-34}$, which is negligible. For the conjectured vacuum-polarization channel, $|\alpha| \sim 10^{-2}$, which would produce percent-level corrections. In the latter case, the effective gravitational coupling of matter is modified from ρ_{m} to $(1 \mp |\alpha|)\rho_{\text{m}}$ (depending on the sign), which is observationally degenerate with a rescaling of Newton's constant at the background level. The correction becomes physically distinct only in the perturbative regime (growth of structure) or in regions with large density contrasts. We discuss the observational consequences in Section 8.

Section Result and Implications

The conceptual framework defines the matter-induced vacuum correction $\Delta\rho_{\text{vac}}(\rho_{\text{m}})$ and states the central hypothesis. The mathematical development that follows will derive this correction from the one-loop effective action.

5 Mathematical Development I: The One-Loop Effective Action and Heat-Kernel Formalism

5.1 The effective action in curved spacetime

Consider a real scalar field ϕ of mass m with a non-minimal coupling ξ to the Ricci scalar, propagating in a curved background spacetime $(M, g_{\mu\nu})$. The classical action is

$$S[\phi, g] = -\frac{1}{2} \int d^4x \sqrt{-g} [g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + (m^2 + \xi R) \phi^2]. \quad (7)$$

The one-loop effective action is obtained by integrating out the quantum fluctuations of ϕ around a classical background:

$$\Gamma_{1\text{-loop}}[g] = \frac{i}{2} \ln \det \left(-\square + m^2 + \xi R \right), \quad (8)$$

where $\square = g^{\mu\nu} \nabla_\mu \nabla_\nu$ is the covariant d'Alembertian [12, 13]. This expression is formally divergent and requires regularization.

5.2 The Schwinger–DeWitt heat-kernel expansion

The standard tool for analyzing the one-loop effective action is the Schwinger–DeWitt proper-time representation [14, 15]. For a second-order differential operator $\hat{F} = -\square + E$, where E is a local potential term (here $E = m^2 + \xi R$), the effective action is represented as

$$\Gamma_{1\text{-loop}} = -\frac{1}{2} \int_0^\infty \frac{ds}{s} \text{Tr} \left(e^{-s\hat{F}} \right), \quad (9)$$

where s is the proper-time parameter. The heat kernel $K(x, x'; s) = \langle x | e^{-s\hat{F}} | x' \rangle$ admits an asymptotic expansion for small s :

$$K(x, x; s) \sim \frac{1}{(4\pi s)^{d/2}} \sum_{n=0}^{\infty} a_n(x) s^n, \quad (10)$$

where d is the spacetime dimension and $a_n(x)$ are the Schwinger–DeWitt (or Seeley–DeWitt–Gilkey) coefficients [15, 16]. These coefficients are local scalar invariants constructed from the curvature, the potential E , and their covariant derivatives.

Proposition 5.1 (Leading heat-kernel coefficients for a massive scalar field). *For the operator $\hat{F} = -\square + m^2 + \xi R$ acting on a scalar field in four-dimensional spacetime, the first three coincidence-limit heat-kernel coefficients are [16, 12]:*

$$a_0 = 1, \quad (11)$$

$$a_1 = \left(\frac{1}{6} - \xi \right) R - m^2, \quad (12)$$

$$\begin{aligned} a_2 = & \frac{1}{180} (R_{\mu\nu\alpha\beta} R^{\mu\nu\alpha\beta} - R_{\mu\nu} R^{\mu\nu}) + \frac{1}{2} \left(\frac{1}{6} - \xi \right)^2 R^2 \\ & + \frac{1}{6} \left(\frac{1}{5} - \xi \right) \square R + \frac{m^4}{2} - m^2 \left(\frac{1}{6} - \xi \right) R. \end{aligned} \quad (13)$$

Proof. The derivation follows from the recursive solution of the transport equations for the heat-kernel coefficients along geodesics, as developed systematically by DeWitt [14] and

refined by Gilkey [17] and Vassilevich [16]. For the operator $\hat{F} = -\square + E$ with $E = m^2 + \xi R$, the coefficients are:

$a_0 = \mathbf{1}$ (the identity on the fiber), which for a scalar field is simply 1.

$a_1 = \frac{1}{6}R - E = \left(\frac{1}{6} - \xi\right) R - m^2$.

$a_2 = \frac{1}{180}(R_{\mu\nu\alpha\beta}R^{\mu\nu\alpha\beta} - R_{\mu\nu}R^{\mu\nu} + \square R) + \frac{1}{2}E^2 - \frac{1}{6}RE + \frac{1}{6}\square E$.

Substituting $E = m^2 + \xi R$ and collecting terms yields (13). The detailed computation is standard; see [16], Theorem 4.1, and [13], Chapter 6. $\square$

5.3 Renormalized effective Lagrangian

After dimensional regularization and renormalization (absorbing the divergent terms into renormalized couplings), the finite part of the one-loop effective Lagrangian contains curvature-dependent contributions. At one-loop order, the renormalized effective Lagrangian for a scalar field of mass m in four dimensions takes the form [12, 13, 15]

$$\mathcal{L}_{\text{eff}} \supset \frac{1}{(4\pi)^2} \left[\frac{m^2}{2} \left(\frac{1}{6} - \xi \right) R \ln \frac{m^2}{\mu^2} + \dots \right], \quad (14)$$

where μ is the renormalization scale and the ellipsis denotes terms involving R^2 , $R_{\mu\nu}R^{\mu\nu}$, and higher-order curvature invariants.

The crucial observation is that *the renormalized effective action depends on curvature invariants*, and these curvature invariants are determined by the matter distribution through Einstein's equations.

Counterarguments and Responses

Objection: The heat-kernel expansion is an asymptotic expansion valid for short proper times (high energies). Can it be trusted for infrared questions like the cosmological constant?

Response: The heat-kernel expansion is used here only to identify the *structure* of the curvature-dependent corrections to the effective action. The specific numerical coefficients (the a_n) are exact for each order. The physically relevant question is whether the leading curvature correction $\propto m^2 R$ contributes meaningfully to the vacuum energy density, and this is determined by the mass spectrum of the fields and the local curvature, not by the convergence properties of the asymptotic series. Moreover, the adiabatic regularization program of Moreno-Pulido and Solà Peracaula [19] provides an independent confirmation of the same curvature-dependent structure using a different regularization scheme.

Objection: The result depends on the renormalization scale μ . What sets μ ?

Response: In the running vacuum framework, the physical renormalization scale is set by the characteristic energy scale of the gravitational background: $\mu \sim H$ (the Hubble rate) for the homogeneous cosmological background, or $\mu \sim \sqrt{R}$ for a locally curved region [19, 18].

This choice is analogous to the standard practice in gauge theories of setting the renormalization scale to the characteristic energy of the process [31].

Section Result and Implications

The one-loop effective action for a quantum field in curved spacetime, computed via the Schwinger–DeWitt heat-kernel expansion, contains curvature-dependent contributions whose leading term is proportional to $m^2 R$. Since curvature is sourced by matter, this establishes the mathematical pathway from matter density to vacuum energy modification.

6 Mathematical Development II: Derivation of the Matter-Dependent Vacuum Correction

6.1 From curvature-dependence to matter-dependence

Consider a region of spacetime containing a non-relativistic matter distribution with energy density ρ_m . In the weak-field, slowly varying regime, the trace of the Einstein equations gives

$$R = -8\pi G T = -8\pi G (T^{(\text{matter})} + T^{(\text{vac})}). \quad (15)$$

For non-relativistic matter, $T^{(\text{matter})} \approx -\rho_m + 3P_m \approx -\rho_m$ (since $P_m \ll \rho_m$). If we work perturbatively around a background where the vacuum contribution to the trace is subdominant, $|T^{(\text{vac})}| \ll |T^{(\text{matter})}|$ (i.e., $\rho_{\text{vac}} \ll \rho_m$ at the scale of interest—a condition that is satisfied in matter-dominated environments but *not* in the present-day cosmic mean, where $\rho_{\text{vac}} \sim \rho_m$), then at leading order

$$R \approx 8\pi G \rho_m. \quad (16)$$

Proposition 6.1 (Matter-dependent vacuum energy from the one-loop effective action). *Consider a single real scalar field of mass m with non-minimal coupling ξ in a background spacetime with Ricci scalar $R = 8\pi G \rho_m$. The one-loop correction to the vacuum energy density, arising from the a_1 heat-kernel coefficient, is*

$$\Delta\rho_{\text{vac}} = -\frac{m^2}{(4\pi)^2} \left(\frac{1}{6} - \xi \right) \cdot 8\pi G \rho_m \cdot \ln \frac{m^2}{\mu^2} \equiv -\alpha_{\text{single}} \rho_m, \quad (17)$$

where

$$\alpha_{\text{single}} = \frac{8\pi G m^2}{(4\pi)^2} \left(\frac{1}{6} - \xi \right) \ln \frac{m^2}{\mu^2}. \quad (18)$$

Proof. The effective Lagrangian from the a_1 coefficient is $\Delta\mathcal{L} = \frac{m^2}{(4\pi)^2} \left(\frac{1}{6} - \xi \right) R \ln \frac{m^2}{\mu^2}$ (from Eq. (14)). The vacuum energy density receives a correction $\Delta\rho_{\text{vac}} = -\Delta\mathcal{L}|_{R=8\pi G \rho_m}$ (the sign

arises from the relation $\rho = -\mathcal{L}$ for a vacuum-type contribution in the convention where $\mathcal{L}$ contributes with a positive sign to the action). Substituting $R = 8\pi G \rho_{\text{m}}$ yields (17). $\square$

6.2 Generalization to the Standard Model field content

The Standard Model contains multiple scalar, fermionic, and vector fields, each contributing to the one-loop effective action. For a collection of N_s real scalar fields, N_f Dirac fermion fields, and N_v vector fields, the total a_1 -induced vacuum correction generalizes to

$$\Delta\rho_{\text{vac}} = - \left[\sum_i \frac{c_i m_i^2}{(4\pi)^2} \left(\frac{1}{6} - \xi_i \right) \ln \frac{m_i^2}{\mu^2} \right] \cdot 8\pi G \rho_{\text{m}} \equiv -\alpha \rho_{\text{m}}, \quad (19)$$

where c_i counts the degrees of freedom with appropriate signs (positive for bosons, negative for fermions due to the fermionic determinant), and ξ_i is the non-minimal coupling for scalars (with appropriate curvature coupling terms for higher-spin fields; see [13, 20] for the spin-1/2 case).

Theorem 6.2 (Linearity of the leading vacuum correction). *Under the following assumptions:*

- (i) *The background spacetime is mildly curved: $R \ll m^2$ for at least one field species with mass m .*
- (ii) *The matter distribution is non-relativistic and slowly varying on the scale of the Compton wavelength m^{-1} : specifically, $|\nabla \rho_{\text{m}}|/(m \rho_{\text{m}}) \ll 1$ and $|\nabla^2 \rho_{\text{m}}|/(m^2 \rho_{\text{m}}) \ll 1$.*
- (iii) *The effective action is computed at one-loop order in the perturbative expansion.*
- (iv) *Self-consistency: the vacuum correction is small compared to the matter density, $|\alpha| \rho_{\text{m}} \ll \rho_{\text{m}}$, i.e., $|\alpha| \ll 1$.*

Then the leading matter-dependent correction to the vacuum energy density is linear in ρ_{m} :

$$\Delta\rho_{\text{vac}} = -\alpha \rho_{\text{m}} + \mathcal{O}\left(\frac{(8\pi G \rho_{\text{m}})^2}{m^4}\right) + \mathcal{O}\left(\frac{\nabla^2 \rho_{\text{m}}}{m^2 \rho_{\text{m}}} \cdot \alpha \rho_{\text{m}}\right), \quad (20)$$

where α is given by Eq. (19). The first error term comes from the a_2 heat-kernel coefficient (quadratic curvature terms); the second comes from derivative terms ($\square R$) in the effective action.

Proof. Under assumption (i), the heat-kernel expansion is organized in powers of $R/m^2 \ll 1$. The leading curvature-dependent term in the effective Lagrangian comes from the a_1 coefficient and is linear in R . Under assumption (iii), higher-loop corrections are suppressed by additional powers of the coupling constants. The a_2 coefficient contributes terms quadratic

in curvature (R^2 , $R_{\mu\nu}R^{\mu\nu}$), which by Eq. (16) are of order $(8\pi G \rho_m)^2/m^4$ relative to the leading term. Assumption (ii) ensures that derivative terms ($\square R \sim 8\pi G \nabla^2 \rho_m$) are suppressed by $(|\nabla^2 \rho_m|)/(m^2 \rho_m)$ relative to the leading correction. Assumption (iv) ensures that the backreaction of the vacuum correction on the geometry does not invalidate the perturbative substitution $R \approx 8\pi G \rho_m$; including the vacuum backreaction would replace ρ_m by $\rho_m + \Delta\rho_{\text{vac}} = (1 - \alpha)\rho_m$ in the trace equation, producing a correction of relative order α that is absorbed into a redefinition of the effective coupling.

The linearization therefore holds provided the matter distribution varies on scales L satisfying $L \gg m^{-1}$ (smoothness condition) and the density satisfies $8\pi G \rho_m \ll m^2$ (weak-curvature condition). For the top quark mass $m_t \sim 173$ GeV, the Compton wavelength is $m_t^{-1} \sim 10^{-18}$ m, and the weak-curvature condition is satisfied for all astrophysical environments below nuclear density ($\rho_m \ll m_t^4 \sim 10^9$ GeV⁴). However, at sharp density boundaries (e.g., halo edges where ρ_m changes on scales $\sim$ kpc), the gradient terms must be estimated explicitly for the specific density profile. $\square$

6.3 Estimation of the coupling coefficient α

We now estimate the magnitude of α for realistic field content. It is essential to distinguish two physically different channels through which the vacuum energy responds to matter, and to be explicit about which is rigorously derived and which is conjectured.

6.3.1 Channel (a): Curvature-dependent effective action (rigorous)

Proposition 6.3 (Rigorous estimate for α from channel (a)). *For the Standard Model field content, the coupling coefficient arising from the curvature-dependent one-loop effective action (i.e., from the heat-kernel a_1 coefficient via $R = 8\pi G \rho_m$) is*

$$\alpha^{(a)} \sim \frac{G m_{\text{eff}}^2}{(4\pi)^2} \sim \frac{m_{\text{eff}}^2}{16\pi^2 M_{\text{Pl}}^2} \sim 10^{-34}, \quad (21)$$

where m_{eff} is an effective mass scale determined by the heaviest particles in the loop. This is negligibly small for all practical purposes.

Proof. From Eq. (18), for a single field:

$$\alpha_{\text{single}} = \frac{8\pi G m^2}{16\pi^2} \left(\frac{1}{6} - \xi \right) \ln \frac{m^2}{\mu^2} = \frac{G m^2}{2\pi} \left(\frac{1}{6} - \xi \right) \ln \frac{m^2}{\mu^2}. \quad (22)$$

Using $G = M_{\text{Pl}}^{-2}/(8\pi)$ in natural units:

$$\alpha_{\text{single}} = \frac{m^2}{16\pi^2 M_{\text{Pl}}^2} \left(\frac{1}{6} - \xi \right) \ln \frac{m^2}{\mu^2}. \quad (23)$$

For the heaviest Standard Model particles ($m \sim m_t \sim 173$ GeV, $M_{\text{Pl}} \sim 2.4 \times 10^{18}$ GeV), this gives $\alpha_{\text{single}} \sim 10^{-34}$, which is negligibly small. This suppression is a consequence of the gravitational weakness: the matter density enters the vacuum energy only through the curvature $R = 8\pi G \rho_{\text{m}}$, and the factor $G \sim M_{\text{Pl}}^{-2}$ suppresses the result. $\square$

6.3.2 Channel (b): Direct vacuum polarization (conjectured)

Channel (a) is not the only mechanism through which matter can modify the vacuum energy. A second, physically distinct channel involves the *direct interaction* of matter fields with vacuum fluctuations through loop diagrams in which the matter distribution provides external sources. We describe this channel and estimate its magnitude, but we emphasize at the outset that *this channel is not rigorously derived from the same heat-kernel formalism as channel (a)*. The estimate below is a conjecture, motivated by standard QFT reasoning but lacking a complete first-principles derivation connecting ρ_{m} as a background energy density in Einstein's equations to ρ_{m} as an external source in vacuum polarization diagrams.

The physical picture is as follows. When matter is present, it contributes to the effective action in two ways: (a) indirectly, by modifying the background curvature $R \sim 8\pi G \rho_{\text{m}}$, which feeds into the vacuum energy through the curvature-dependent terms in the heat-kernel expansion; and (b) directly, through vacuum polarization diagrams in which the matter fields run in the loop and the background matter distribution provides external sources.

Channel (a) produces the gravitationally suppressed corrections derived above. Channel (b), if operative, involves the full strength of the gauge and Yukawa couplings.

Conjecture 6.4 (Vacuum polarization contribution). *If the background matter distribution acts as an effective external source for vacuum polarization loops, the resulting correction to the vacuum energy density is of order*

$$|\Delta\rho_{\text{vac}}^{(\text{VP})}| \sim \frac{g^2}{16\pi^2} N_c \rho_{\text{m}} \cdot f(m_f/\mu), \quad (24)$$

where g is the gauge coupling, N_c the color factor, $f(m_f/\mu)$ a dimensionless function of order unity containing logarithmic dependence on m_f/μ , and the sign is not determined by the present analysis. For the strong interaction ($\alpha_s \approx 0.12$) with $N_c = 3$, or for the top Yukawa coupling ($y_t \approx 1$), this yields $|\alpha^{(b)}| \sim 10^{-2}$.

The vacuum polarization tensor $\Pi_{\mu\nu}(q)$ for a fermion loop in a gauge field background has the standard form [30]

$$\Pi_{\mu\nu}(q) = (q^2 g_{\mu\nu} - q_\mu q_\nu) \Pi(q^2), \quad (25)$$

with

$$\Pi(q^2) = -\frac{g^2 N_c}{16\pi^2} \left[\frac{1}{3} \ln \frac{m_f^2}{\mu^2} + \text{finite terms} \right]. \quad (26)$$

The crucial step—how a background matter stress-energy tensor couples to these vacuum polarization loops to produce an energy density correction proportional to ρ_m —is the gap in the present analysis. In standard flat-space QFT, the vacuum polarization modifies the photon propagator in the presence of external charges. The analogy would be that the background matter distribution, acting through its gravitational and gauge interactions, modifies the vacuum fluctuation spectrum in a way that produces an effective energy density shift proportional to ρ_m . However, making this precise requires a calculation that has not been carried out in the literature, and the estimate (24) should therefore be regarded as a conjecture whose verification or refutation is an important task for future work.

6.3.3 Summary: two regimes

The situation is as follows. The rigorously derived result from channel (a) gives $|\alpha^{(a)}| \sim 10^{-34}$ —too small to be observable. The conjectured channel (b) gives $|\alpha^{(b)}| \sim 10^{-2}$ —potentially observable. *The central quantitative claim of this paper is that channel (b) exists and is of the estimated magnitude, but this claim has the status of a conjecture, not a theorem.* The structural result—that matter-sourced curvature produces a matter-dependent vacuum correction (Theorem 6.2)—is rigorous, but the magnitude that would make the effect observationally relevant depends on the conjectured channel.

Counterarguments and Responses

Objection: The vacuum polarization mechanism described here is standard QFT. Why has it not been identified before as a source of matter-dependent vacuum energy?

Response: The individual ingredients—vacuum polarization, curvature-dependent effective action, matter-sourced curvature—are indeed well known. What has not been explicitly developed in the literature is the chain of reasoning that connects them to a *local* matter-dependent vacuum energy density. Most treatments of the vacuum energy in curved space-time work on homogeneous (FLRW) backgrounds, where the matter density is a function of time only. The present analysis extends the logic to inhomogeneous configurations, where ρ_m varies in space. The running vacuum model of Solà Peracaula [18, 19] captures the time-dependence; the present paper argues that the spatial dependence follows from the same physics.

Objection: Channel (b) is not derived from the heat-kernel formalism. How can it be presented alongside the rigorous channel (a)?

Response: The referee is correct that channels (a) and (b) rest on different levels of rigor, and the revised paper now states this explicitly. Channel (a) is a rigorous consequence of the Schwinger–DeWitt expansion. Channel (b) is a conjecture motivated by standard vacuum-polarization physics, whose verification requires a calculation that connects the matter stress-energy tensor to external sources in polarization loops. We present both because

they illustrate the two physically distinct mechanisms, and because the conjectured channel identifies a concrete target for future first-principles computation. The paper’s structural result (Theorem 6.2) is independent of channel (b).

Section Result and Implications

The one-loop effective action in curved spacetime, combined with the Einstein constraint $R \sim 8\pi G \rho_m$, produces a rigorously derived matter-dependent correction to the vacuum energy density with $|\alpha^{(a)}| \sim 10^{-34}$ (channel a). A conjectured but physically motivated vacuum-polarization mechanism (channel b) suggests $|\alpha^{(b)}| \sim 10^{-2}$. The structural result—linearity of the leading correction (Theorem 6.2)—is rigorous; the observationally relevant magnitude depends on the conjectured channel.

7 Mathematical Development III: Connection to Running Vacuum and Brown’s QKDE Frameworks

7.1 Structural analogy with the running vacuum model

The running vacuum model (3) expresses the vacuum energy density as a function of the Hubble rate:

$$\rho_{\text{vac}}(H) = \rho_{\text{vac}}^{(0)} + \frac{3\nu}{8\pi} M_{\text{Pl}}^2 H^2 + \dots \quad (27)$$

On a homogeneous background, the Friedmann equation gives $H^2 = (8\pi G/3) \rho_m$ (in the matter-dominated era, neglecting radiation and vacuum contributions at leading order). Substituting:

$$\rho_{\text{vac}}(H) = \rho_{\text{vac}}^{(0)} + \frac{3\nu}{8\pi} M_{\text{Pl}}^2 \cdot \frac{8\pi G}{3} \rho_m = \rho_{\text{vac}}^{(0)} + \nu \rho_m. \quad (28)$$

Proposition 7.1 (RVM–matter-density correspondence). *On a homogeneous matter-dominated background, the running vacuum model with parameter ν is equivalent to the matter-dependent vacuum energy model with coupling $\alpha = -\nu$. Since the running vacuum model predicts $\nu > 0$ (with $|\nu| \sim 10^{-3}$ [21]), this corresponds to $\alpha < 0$, i.e., the vacuum energy increases with matter density in the RVM.*

Proof. Comparing Eq. (28) with the ansatz $\rho_{\text{vac}}(\rho_m) = \rho_{\text{vac}}^0 + \Delta\rho_{\text{vac}}(\rho_m)$ yields $\Delta\rho_{\text{vac}} = \nu \rho_m$, i.e., $\alpha = -\nu$. The sign of ν is determined by the field content through the adiabatic regularization calculation of [19]. $\square$

Remark 7.2 (On the sign of α). The sign of the matter-dependent correction is a subtle and important question. In the phenomenological model of Paper 2 [29], $\alpha > 0$ was assumed (vacuum energy suppressed by matter). The RVM calculation gives $\nu > 0$, which through the above correspondence yields $\alpha < 0$ (vacuum energy enhanced by matter density on a

homogeneous background). However, this sign discrepancy may be resolved by noting that the RVM result pertains to the *homogeneous* background, where increasing ρ_m corresponds to earlier cosmic epochs with higher H . In the *inhomogeneous* case—a local overdensity embedded in the cosmic background—the relevant curvature includes both the background FLRW contribution and the perturbation, and the sign of the vacuum correction depends on the specific channel (gravitational vs. vacuum-polarization) and the field content. The physical question of the sign is ultimately empirical and must be settled by confrontation with data.

7.2 Structural analogy with Brown’s QKDE framework

The following connection is noted as the present author’s interpretive observation, not as a claim of Brown’s original paper [26, 27]. Brown’s original motivation for the curvature-motivated kinetic normalization concerns the dynamics of the kinetic sector—specifically, wave-function renormalization of a light scalar by heavy fields generating dimension-six operators $\sim (R/M^2)X$ [26]—not vacuum energy running or local matter-density dependence directly. Nevertheless, the mathematical structure of Brown’s QKDE framework admits a structural analogy with the matter-dependent vacuum model developed in this paper, as we now show.

In Brown’s QKDE framework [26], the curvature-motivated kinetic normalization

$$K = 1 + \frac{\alpha_K R}{M^2} \quad (29)$$

modifies the scalar energy density by a term $\delta\rho_\phi \sim (\alpha_K R/M^2) \cdot \dot{\phi}^2/2$. On a background where $R \sim 8\pi G \rho_m$, this gives

$$\delta\rho_\phi \sim \frac{8\pi G \alpha_K}{M^2} \rho_m \cdot \frac{\dot{\phi}^2}{2} \propto \rho_m. \quad (30)$$

Proposition 7.3 (QKDE–matter-density structural analogy (present author’s observation)). *In Brown’s QKDE framework [26] with curvature-motivated kinetic normalization, the scalar dark-energy sector acquires a correction to its energy density that is linear in the local matter density. This correction has the same structural form as the matter-dependent vacuum energy model (6). This structural analogy is identified by the present author; it is not part of Brown’s original construction, which addresses the dynamics of the kinetic sector rather than local vacuum-energy dependence on matter density [26, 27].*

Proof. The scalar energy density in Brown’s QKDE framework is $\rho_\phi = K\dot{\phi}^2/2 + V(\phi)$. With

K given by (29) and $R \approx 8\pi G \rho_{\text{m}}$ at leading order:

$$\rho_{\phi} = \left(1 + \frac{8\pi G \alpha_K \rho_{\text{m}}}{M^2}\right) \frac{\dot{\phi}^2}{2} + V(\phi) = \rho_{\phi}^{(0)} + \frac{8\pi G \alpha_K \dot{\phi}^2}{2M^2} \rho_{\text{m}}, \quad (31)$$

where $\rho_{\phi}^{(0)} = \dot{\phi}^2/2 + V(\phi)$ is the scalar energy density in the absence of the curvature correction. This is a linear correction in ρ_{m} with coefficient $\alpha_{\text{QKDE}} = 8\pi G \alpha_K \dot{\phi}^2/(2M^2)$. It is important to emphasize—as also noted in the companion Paper 1a [28]—that this is a structural analogy, not an identification: Brown’s QKDE framework and the present matter-dependent vacuum model share an R -dependent modification of the dark-energy sector but differ in their physical origin (kinetic-sector dynamics vs. vacuum polarization), their perturbative structure ($c_s^2 = 1$ identically in QKDE, while the present model does not introduce a propagating scalar), and their relationship to the cosmological constant. $\square$

7.3 Unified interpretation (present author’s synthesis)

The following synthesis is the present author’s interpretive observation. The three approaches—the microscopic QFT derivation (Section 6), the running vacuum model, and Brown’s QKDE framework [26]—can be seen as converging on the same qualitative conclusion: the effective vacuum (or dark-energy) sector acquires a correction that is linear in the local matter density at leading order. Brown’s framework was not designed to address local matter-density dependence; the connection drawn here is a structural analogy identified by the present author.

Remark 7.4 (Structural analogy across frameworks). Three independent theoretical frameworks—one-loop QFT in curved spacetime (Theorem 6.2), the running vacuum model [19, 21], and Brown’s QKDE construction [26]—all yield, after framework-specific substitutions ($R \propto \rho_{\text{m}}$ for the heat-kernel approach, $H^2 \propto \rho_{\text{m}}$ for the RVM, $R \propto \rho_{\text{m}}$ for QKDE), an effective vacuum or dark-energy density that contains a term linear in ρ_{m} at leading order. The three frameworks differ in their detailed assumptions, their perturbative structures, the physical identity of the corrected quantity (vacuum energy vs. scalar dark energy), and even the sign and magnitude of the coupling coefficient, but they share the same structural form:

$$\rho_{\text{vac}}^{\text{eff}} = \rho_{\text{vac}}^{(0)} + \beta \rho_{\text{m}} + \mathcal{O}(\rho_{\text{m}}^2), \quad (32)$$

where β is a small dimensionless coefficient whose sign and magnitude depend on the specific framework and the field content. This structural analogy does not constitute a proof of convergence—the framework-specific substitutions operate on different backgrounds and carry different physical content—but it suggests that the matter-linear correction is a robust structural feature of QFT in curved spacetime rather than an artifact of a particular formulation.

Argument. For the one-loop QFT framework, the linear dependence on ρ_m is established by Theorem 6.2. For the RVM, it follows from Proposition 7.1 via the substitution $H^2 \propto \rho_m$ (valid on a matter-dominated homogeneous background). For Brown’s QKDE framework, it follows from Proposition 7.3 (the present author’s structural analogy). The three derivations start from different premises (heat-kernel expansion, adiabatic regularization, and an EFT action principle, respectively) and use different mathematical tools.

Counterarguments and Responses

Objection: The three frameworks give different values and even different signs for the coupling coefficient. How can they be said to “converge”?

Response: The convergence is structural, not quantitative. All three frameworks predict a leading-order linear correction in ρ_m ; the quantitative details depend on the specific field content, the regularization scheme, and the physical channel (gravitational vs. matter-coupling). The structural convergence is analogous to the situation in perturbative QCD, where different regularization schemes give different numerical values for loop integrals but agree on the α_s -dependence of physical observables. The sign ambiguity noted in Remark 7.2 is a genuine open question that requires either a first-principles calculation with specified field content or empirical determination.

Objection: Brown’s QKDE framework modifies the dark-energy sector, not the vacuum energy. Is it legitimate to identify the two?

Response: This objection touches the heart of the separation thesis. In Brown’s QKDE framework [26], the “dark energy” is carried by a scalar sector that is *distinct* from the vacuum energy—and Brown’s construction was motivated by kinetic-sector dynamics, not by vacuum energy considerations [26, 27]. The structural analogy (Proposition 7.3), identified by the present author, shows that both quantities respond to curvature (and hence to matter density) in a structurally similar way. Whether one calls this response a “vacuum energy correction” or a “scalar dark-energy correction” is a question of interpretation that goes beyond Brown’s original framework. The physically observable quantity is the total effective dark-energy density, which is what the model predicts.

Section Result and Implications

The matter-dependent vacuum energy model derived in Section 6 is structurally consistent with the running vacuum model and—as the present author’s interpretive observation—exhibits a structural analogy with Brown’s QKDE framework [26]. All three frameworks predict a linear matter-density correction at leading order. This convergence from independent theoretical starting points strengthens the case that such a correction is a generic feature of QFT in curved spacetime, though it should be noted that the QKDE connection is an analogy rather than a formal identification.

8 Empirical and Data-Grounded Illustrations

8.1 Modified Friedmann equation

In the presence of a matter-dependent vacuum energy, the Friedmann equation on a homogeneous background becomes

$$H^2 = \frac{8\pi G}{3} [\rho_m + \rho_{\text{vac}}(\rho_m)] = \frac{8\pi G}{3} [(1 - \alpha)\rho_m + \rho_{\text{vac}}^0], \quad (33)$$

where we have used $\rho_{\text{vac}}(\rho_m) = \rho_{\text{vac}}^0 - \alpha \rho_m$.

Proposition 8.1 (Degeneracy with rescaled Newton’s constant). *At the background level, the modified Friedmann equation (33) is equivalent to the standard Friedmann equation with an effective Newton’s constant $G_{\text{eff}} = (1 - \alpha)G$ and a standard cosmological constant $\Lambda_{\text{eff}} = 8\pi G \rho_{\text{vac}}^0$.*

Proof. Equation (33) can be rewritten as

$$H^2 = \frac{8\pi G_{\text{eff}}}{3} \rho_m + \frac{\Lambda_{\text{eff}}}{3}, \quad (34)$$

with $G_{\text{eff}} = (1 - \alpha)G$ and $\Lambda_{\text{eff}} = 8\pi G \rho_{\text{vac}}^0$. At the background level, where ρ_m is spatially homogeneous, this is indistinguishable from the standard Λ CDM model with rescaled parameters. $\square$

This degeneracy means that the model is automatically consistent with background-level cosmological observations (CMB distance measures, BAO, SNe Ia), provided the effective parameters are calibrated appropriately.

8.2 Breaking the degeneracy: perturbation theory

A note on the physical nature of the correction. An important distinction must be made regarding how the vacuum correction enters the perturbation equations. If the correction $\Delta\rho_{\text{vac}} = -\alpha \rho_m$ behaves as a vacuum-type contribution ($w = -1$, entering with $p_{\text{vac}} = -\Delta\rho_{\text{vac}}$), it modifies the background expansion but does not cluster at linear order; it therefore affects the growth equation only through $H(a)$, not through a direct source term. If instead the correction enters as a modification of the effective matter coupling (a rescaling of the gravitational response of matter), the perturbation equations are modified differently. The present analysis does not fully resolve this distinction, which depends on how the vacuum correction responds to perturbations—a question that requires specifying the full perturbed stress-energy tensor for the vacuum sector. The preliminary estimates below assume the simpler case of a background-level modification, acknowledging that a complete perturbation analysis is needed.

The degeneracy is broken at the perturbative level. In regions of different matter density (clusters vs. voids), the effective vacuum energy is different. This produces two potentially testable effects:

First, the growth rate of structure is modified. If the vacuum correction enters primarily through the background expansion rate, the linear growth factor $D(a)$ satisfies a modified equation:

$$D'' + \left(2 + \frac{H'}{H}\right) D' - \frac{3}{2} \Omega_m^{\text{eff}}(a) D = 0, \quad (35)$$

where $\Omega_m^{\text{eff}} = (1 - \alpha) \Omega_m$ and primes denote $d/d \ln a$. For $|\alpha| \sim 10^{-2}$ (the conjectured channel b magnitude), this would produce percent-level modifications to the growth rate $f\sigma_8(z)$. However, this estimate has not been verified by numerically solving the growth equation with the modified background, and the actual magnitude depends on whether the vacuum correction clusters or remains homogeneous—a question that is not settled in the present analysis.

Second, gravitational lensing in overdense regions could be affected because the effective dark-energy density is locally modified. The lensing potential $\Phi + \Psi$ would acquire a correction proportional to $\alpha \delta\rho_m$, where $\delta\rho_m$ is the matter density perturbation, provided the vacuum correction responds to density perturbations (not only to the background density).

8.3 Observational constraints

Current observational bounds on deviations from Λ CDM at the percent level come from several sources. Planck CMB data [33] constrain the total matter content to $\Omega_m h^2 = 0.1430 \pm 0.0011$, and the growth rate measurements from redshift-space distortions give $f\sigma_8(z = 0.38) = 0.497 \pm 0.045$ [21]. The DESI 2024 BAO data release [34] provides additional constraints on the dark-energy equation of state. These data are consistent with Λ CDM but do not exclude percent-level deviations of the type predicted by the matter-dependent vacuum model.

A dedicated Bayesian analysis confronting the modified Friedmann equation (33) with CMB, BAO, and SNe Ia data is needed to constrain α directly. Such an analysis is beyond the scope of the present paper but is planned as part of the research program (Paper #7).

Counterarguments and Responses

Objection: If the model is degenerate with Λ CDM at the background level, it is not falsifiable from background observations alone.

Response: Correct. Falsifiability requires perturbation-level tests. The model predicts specific modifications to the growth rate and lensing signal that are in principle measurable by current and forthcoming surveys (Euclid, Rubin/LSST, DESI). The key observable is the *environment-dependence* of the effective dark-energy density: the model predicts that void

regions have higher effective vacuum energy than cluster regions. This is a qualitatively distinct prediction from Λ CDM, where the dark-energy density is spatially uniform.

Section Result and Implications

The matter-dependent vacuum model is consistent with background-level cosmological observations by construction (through a degeneracy with rescaled Newton’s constant). It becomes potentially distinguishable at the perturbative level, but the precise observational signatures depend on whether the vacuum correction enters as a $w = -1$ background contribution or as a modification of the matter coupling—a question requiring a full perturbation analysis beyond the scope of this paper. If the conjectured channel (b) is operative with $|\alpha| \sim 10^{-2}$, the modifications could be at the percent level; if only the rigorous channel (a) contributes, the effect is negligible.

9 Discussion

The central result of this paper is that the effective vacuum energy density acquires a dependence on the local matter density through standard mechanisms of one-loop QFT in curved spacetime. This dependence is not an exotic or speculative proposal but a consequence of the fact that (a) the renormalized effective action depends on curvature invariants (established by the Schwinger–DeWitt formalism), and (b) curvature is sourced by matter (established by Einstein’s equations).

The physical interpretation is straightforward: matter modifies the boundary conditions and the fluctuation spectrum of quantum fields, analogous to how conducting plates modify the electromagnetic vacuum in the Casimir effect. The renormalized vacuum energy density therefore becomes an environment-dependent quantity.

Several aspects of the result deserve emphasis.

First, the linearity of the leading correction (Theorem 6.2) is not an assumption but a consequence of the perturbative expansion of the heat-kernel coefficients in powers of curvature. Higher-order corrections, proportional to ρ_m^2 and higher, arise from the a_2 and higher coefficients and from multi-loop diagrams.

Second, the magnitude of the coupling depends critically on which physical channel dominates. The rigorously derived channel (a) gives $|\alpha^{(a)}| \sim 10^{-34}$ —negligible. The conjectured channel (b), involving direct vacuum polarization at gauge/Yukawa coupling strength, gives $|\alpha^{(b)}| \sim 10^{-2}$ —potentially observable. The gap between these estimates is the paper’s central open problem. If channel (b) can be established through a first-principles calculation, the model makes definite predictions; if it cannot, the structural result (Theorem 6.2) remains valid but the effect is astrophysically irrelevant. This honest assessment is in sharp contrast to the cosmological constant problem itself, where the discrepancy is 120 orders of

magnitude.

Third, the structural analogy across three independent frameworks—one-loop QFT, running vacuum, and Brown’s QKDE [26]—identified in Remark 7.4, suggests that the matter-linear correction is a robust structural feature. As emphasized in Section 7, the connection to QKDE is the present author’s structural analogy, not a feature of Brown’s original construction; nonetheless, the mathematical correspondence is nontrivial. Each framework employs different mathematical tools and makes different approximations, yet all yield a linear matter-density correction at leading order.

Fourth, the model does not claim to solve the cosmological constant problem. The asymptotic vacuum energy density ρ_{vac}^0 remains a free parameter whose smallness is not explained. What the model does is to provide a microscopic basis for the *matter-dependence* of the vacuum contribution, which is a separate physical question from the value of ρ_{vac}^0 itself. As argued in the companion Paper 1a [28], the separation thesis dissolves the cosmological constant problem by distinguishing between the geometric cosmological constant Λ and the quantum vacuum energy density ρ_{vac} ; the present paper contributes a microscopic mechanism for how ρ_{vac} depends on its environment.

Counterarguments and Responses

Objection: The model assumes that vacuum energy “gravitates” in the usual sense (enters the right-hand side of Einstein’s equations). But the Rugh–Zinkernagel critique [7] questions precisely this assumption.

Response: The present model is compatible with both positions. If vacuum energy gravitates in the standard semiclassical sense, then the matter-dependent correction modifies the effective gravitational source. If it does not gravitate (or gravitates differently), then the correction is physically irrelevant at the gravitational level, and the cosmological constant problem dissolves for different reasons. In either case, the *existence* of the matter-dependent correction in the effective action is a mathematical fact of QFT in curved spacetime; its gravitational *significance* depends on the semiclassical coupling prescription.

Objection: The estimate of α from channel (b) (vacuum polarization) involves gauge and Yukawa couplings that are well-constrained by particle physics. Can the model survive precision tests from collider data?

Response: The vacuum polarization effects discussed here are already included in the Standard Model radiative corrections. They are not “new” effects but a reinterpretation of how known effects contribute to the vacuum energy in curved spacetime. Collider experiments probe vacuum polarization through its effects on scattering amplitudes; the gravitational effects of the same diagrams are suppressed by the weakness of gravity and are not accessible to colliders. The two domains are complementary, not competing.

Objection: If the effective vacuum energy depends on local matter density, does this

introduce an environment-dependent contribution to gravitational dynamics that violates the strong equivalence principle (SEP)?

Response: This is an important foundational question. The Casimir effect provides a useful precedent: the vacuum energy between conducting plates depends on the boundary conditions (plate separation), yet the Casimir force is fully consistent with general relativity—it arises from the stress-energy tensor of the quantum field in the given geometry and does not violate the equivalence principle [37]. In the present model, the vacuum correction $\Delta\rho_{\text{vac}}(\rho_{\text{m}})$ arises from the curvature-dependent effective action, which is itself a consequence of QFT on a curved background consistent with GR. The correction does not introduce a new “fifth force” but modifies the effective stress-energy that sources gravity. However, there is a subtlety: if the vacuum energy varies spatially (because ρ_{m} varies spatially), its gradient contributes an effective pressure gradient that could, in principle, produce testable deviations from the predictions of GR with a constant cosmological term. Whether this constitutes a violation of the SEP depends on whether one defines the SEP to include the constancy of Λ ; in the present framework, Λ remains constant (as in unimodular gravity), and the spatial variation is in the quantum vacuum contribution, which is a matter-sector quantity. A detailed analysis of the SEP implications—including potential tests via lunar laser ranging or pulsar timing—is deferred to future work.

Section Result and Implications

The matter-dependent vacuum energy model is a structural consequence of QFT in curved spacetime (Theorem 6.2), and the structural analogy across multiple frameworks (Remark 7.4) suggests the result is robust. The observationally relevant magnitude ($|\alpha| \sim 10^{-2}$) depends on the conjectured vacuum-polarization channel, which requires further theoretical development.

10 Limitations

This paper is subject to several limitations that should be clearly acknowledged.

First, and most importantly, the central quantitative claim—that the matter-dependent vacuum correction could reach $|\alpha| \sim 10^{-2}$ —rests on the conjectured vacuum-polarization channel (channel b), which is not derived from the same heat-kernel formalism as the rigorous channel (a). The gap between the two channels (Section 6) is the paper’s critical weakness. The rigorous result ($|\alpha^{(a)}| \sim 10^{-34}$) is too small to be observationally relevant; the conjectured result ($|\alpha^{(b)}| \sim 10^{-2}$) is potentially observable but requires a first-principles derivation that has not been provided (see Appendix B).

Second, the sign of α is not uniquely determined by the present analysis (see Remark 7.2). The phenomenological model of Paper 2 [29] assumes $\alpha > 0$ (vacuum suppression), while

the RVM correspondence (Proposition 7.1) suggests $\alpha < 0$ (vacuum enhancement) on a homogeneous background. The paper now presents both signs symmetrically, but a model that cannot predict the sign of its only free parameter has limited theoretical content until this ambiguity is resolved.

Third, the derivation in Section 6 is perturbative and assumes weak curvature ($R \ll m^2$) and slowly varying matter distributions ($|\nabla \rho_m|/(m \rho_m) \ll 1$). In regions of extreme density (neutron stars, black hole environments), the perturbative expansion breaks down. At sharp density boundaries (halo edges), gradient terms in the heat-kernel expansion ($\square R$ terms) could dominate over the ρ_m^2 corrections, as noted in Theorem 6.2.

Fourth, all results pertain to the one-loop level. Higher-loop corrections, non-perturbative effects, and possible resummations are beyond the scope of the present analysis.

Fifth, the connection to Brown’s QKDE framework (Section 7) is a structural analogy identified by the present author, not a formal derivation from Brown’s framework. Brown’s QKDE framework [26] operates at the level of an EFT action principle and was motivated by kinetic-sector dynamics (wave-function renormalization), not by the local matter-density dependence developed in this paper [27]. A formal embedding of the matter-dependent vacuum model within the QKDE framework would require specifying the relationship between the kinetic normalization $K(\chi)$ and the vacuum polarization effects, and such an embedding has not been established.

Sixth, the observational analysis in Section 8 is preliminary. The perturbation analysis does not specify whether the vacuum correction enters as a $w = -1$ background contribution or as a modification of the matter coupling, and the growth-rate estimate has not been verified by numerically solving the modified growth equation. A full Bayesian comparison with cosmological data is needed.

Seventh, this paper references several other papers in the author’s research program (Papers #1a, #2, #4) whose foundational claims are not independently verifiable by the reader. The present paper is intended to be self-contained in its mathematical development; the program references provide context but are not logically required for the arguments presented here.

11 Future Directions

The results of this paper suggest several directions for future work.

(1) *First-principles derivation of channel (b)*. The most urgent open problem is to derive (or refute) the conjectured vacuum-polarization channel from first principles—specifically, to compute the renormalized stress-energy tensor of a fermion field in the presence of both a curved background and a classical matter source, and to determine whether the matter density enters the vacuum energy correction with gauge-coupling strength ($g^2/(16\pi^2)$) rather than gravitational strength ($Gm^2/(16\pi^2)$).

(2) *Sign determination.* Resolving the sign of α through explicit calculation (even for a simplified field content such as a single massive fermion) or through confrontation with growth-rate data ($f\sigma_8$) from DESI, Euclid, and Rubin/LSST.

(3) *Full perturbation analysis.* Specifying whether the vacuum correction enters as a $w = -1$ contribution or as a matter-coupling modification, and numerically solving the growth equation to obtain quantitative predictions for $f\sigma_8(z)$ and lensing observables.

(4) *Non-linear regime.* Extending the analysis to the non-linear regime (high-density regions, virialized halos) would enable predictions for galaxy rotation curves and cluster dynamics.

(6) *Formal embedding in Brown’s QKDE framework.* Establishing a formal connection between the vacuum polarization mechanism and Brown’s kinetic normalization [26] would unify the microscopic and EFT-level descriptions.

(7) *UV completion.* Investigating whether a UV-complete theory of quantum gravity (e.g., asymptotic safety [35], loop quantum gravity) predicts or constrains the matter-dependent vacuum correction.

12 Position Within the Research Program

This article is Paper #3 in the research program “What If the Vacuum Gravitates Locally? Separating Cosmic Expansion from Quantum Vacuum Energy,” developed at the Institute of Integrative and Interdisciplinary Research (Toronto) and the Information Physics Institute (Gosport). As indicated in the program listing of Paper 1a [28], the present paper belongs to Part II (Theoretical Microphysics) and carries the subtitle “Towards a Quantum Field Theory Derivation.”

Its role is to provide the microscopic QFT foundation for the phenomenological matter–vacuum coupling introduced in Paper #4 [29] and to connect this coupling to the separation thesis established in Paper #1a [28]. Specifically, it demonstrates that the dependence $\rho_{\text{vac}} = \rho_{\text{vac}}(\rho_{\text{m}})$ is not an ad hoc assumption but a consequence of standard one-loop QFT in curved spacetime.

The present paper draws on Brown’s QKDE framework [26], integrated into the program through Paper #1a, to argue—as the present author’s interpretive observation—that the matter-dependent correction is structurally analogous to features of a GR-preserving EFT of dark energy. Future papers in the program—particularly Paper #5 (sequestering mechanism), Paper #7 (structure growth), and Paper #12 (vacuum capture model)—will build on the results established here.

Section Result and Implications

The present paper provides the microscopic theoretical basis for the research program’s central hypothesis. By grounding the matter–vacuum coupling in established QFT mechanisms, it connects the phenomenological papers (Papers #2, #4) to the foundational review (Paper #1a) and opens the pathway to observational tests (Papers #7, #8, #16).

13 Recommended Journal for Publication

This article is recommended for submission to **Foundations of Physics** (Springer).

The recommendation is based on the following considerations. The manuscript develops a microscopic QFT mechanism for the dependence of vacuum energy on matter density, a topic that sits at the intersection of quantum field theory, general relativity, and the philosophy of fundamental physics. *Foundations of Physics* has a long history of publishing work that critically examines the conceptual and mathematical underpinnings of physical theories. The journal’s readership spans cosmology, high-energy theory, condensed-matter physics, and philosophy of science—precisely the disciplines engaged by the present paper.

The mathematical content (propositions, theorems, heat-kernel formalism, connections to the RVM and Brown’s QKDE framework) is rigorous but accessible, consistent with the journal’s standard. The falsifiable predictions (Section 8) connect the foundational analysis to ongoing observational programs.

Alternative venues include *Classical and Quantum Gravity* (IOP), *European Physical Journal C* (Springer), and *Physical Review D* (APS), depending on the emphasis desired.

Acknowledgments

The author is grateful to Daniel Brown (University of Utah) for developing the Quantum-Kinetic Dark Energy framework. Brown’s QKDE paper [26]—a comprehensive, self-contained construction published as open access in the *International Journal of Modern Physics D*—brings substantial clarity to the effective-field-theory landscape of dark energy. In personal communication [27], Brown generously encouraged the author to reference and build upon the QKDE formalism. The structural analogies drawn between Brown’s framework and the matter-dependent vacuum model in Section 7 are the present author’s interpretive observations; Brown’s original motivation concerns the dynamics of the kinetic sector, not vacuum energy running or local matter-density dependence. Any errors in the presentation of QKDE-related results are solely the responsibility of the present author.

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A Heat-Kernel Coefficients for Higher-Spin Fields

For completeness, we collect the heat-kernel coefficients relevant to the Standard Model field content, following [16, 13].

Spin-0 (real scalar): The coefficients a_0 , a_1 , a_2 are given in Proposition 5.1.

Spin-1/2 (Dirac fermion): For a Dirac fermion of mass m_f in four dimensions, the operator is $\hat{F} = -\not{D}^2 + m_f^2$, where $\not{D}$ is the Dirac operator in curved spacetime. The leading coefficients are [13]:

$$a_0^{(1/2)} = 4, \tag{36}$$

$$a_1^{(1/2)} = -\frac{1}{3}R - 4m_f^2, \tag{37}$$

$$a_2^{(1/2)} = \frac{7}{360}R_{\mu\nu\alpha\beta}R^{\mu\nu\alpha\beta} + \frac{1}{36}R_{\mu\nu}R^{\mu\nu} - \frac{1}{30}\square R + 8m_f^4 + \frac{2}{3}m_f^2R, \tag{38}$$

where the factor of 4 in $a_0^{(1/2)}$ counts the four spinor components.

Spin-1 (massive vector, Proca): For a massive vector field of mass m_V , the leading coefficients are [16]:

$$a_0^{(1)} = d = 4, \tag{39}$$

$$a_1^{(1)} = \frac{d-6}{6}R - dm_V^2 = -\frac{1}{3}R - 4m_V^2. \tag{40}$$

Note that fermionic contributions enter with a relative minus sign in the effective action due to the Grassmann nature of the fermion fields: $\Gamma_{1\text{-loop}}^{(1/2)} = -i \ln \det(\hat{F}_{1/2})$.

B On the Vacuum Polarization Channel: Derivation Status and Open Questions

This appendix elaborates on the conjectured vacuum-polarization channel (channel b) discussed in Section 6, and explicitly identifies the steps that constitute rigorous results versus those that remain conjectural.

Step 1 (rigorous). Consider a massive fermion field ψ of mass m_f interacting with a $U(1)$ gauge field A_μ with coupling g in a curved background $(M, g_{\mu\nu})$. The one-loop effective

action receives a contribution from the fermion determinant:

$$\Gamma_{\text{VP}} = -i \ln \det(i\mathcal{D} - m_f), \quad (41)$$

where $\mathcal{D} = \gamma^\mu(\nabla_\mu - igA_\mu)$. The vacuum polarization tensor in flat space is [30]:

$$\Pi(q^2) = -\frac{g^2 N_c}{16\pi^2} \left[\frac{1}{3} \ln \frac{m_f^2}{\mu^2} + \int_0^1 dx \, 2x(1-x) \ln \left(1 - \frac{q^2 x(1-x)}{m_f^2} \right) \right]. \quad (42)$$

At zero momentum transfer, $\Pi(0) = -g^2 N_c / (48\pi^2) \cdot \ln(m_f^2/\mu^2)$.

Step 2 (rigorous in flat space). In flat-space QFT with an external classical source J_μ coupled to the gauge field, the vacuum energy receives a correction from the one-loop self-energy evaluated on the source background. The energy density shift is given by the expectation value of the renormalized stress-energy tensor in the presence of the source [30].

Step 3 (conjectural). The identification of ρ_m (a gravitational source, entering through the Einstein equations) with an effective external source for the vacuum polarization diagrams is the step that lacks a rigorous derivation. In the heat-kernel formalism (channel a), the matter density enters through $R = 8\pi G \rho_m$, and the coupling to vacuum fluctuations is gravitational, yielding $\alpha \sim Gm^2/(16\pi^2)$. For channel (b) to produce $\alpha \sim g^2/(16\pi^2)$, one would need to show that the matter stress-energy tensor, through its gauge and Yukawa interactions with the quantum fields, acts as an effective source that modifies the vacuum polarization diagrams with coupling strength g rather than $\sqrt{G}$.

A plausible physical mechanism is the following: in a region with matter density ρ_m , the fermion propagator is modified by the mean-field background of the matter distribution (e.g., through Hartree-type corrections), which shifts the effective mass and coupling of the fermions in the loop. The resulting energy density correction would be

$$\Delta\rho_{\text{vac}}^{(\text{VP})} \sim -\frac{g^2 N_c}{48\pi^2} \rho_m \ln \frac{m_f^2}{\mu^2} + \mathcal{O}(g^4), \quad (43)$$

where the factor $1/(48\pi^2) = 1/(3 \cdot 16\pi^2)$ combines the $1/3$ from the polarization tensor at zero momentum with the standard loop factor. However, deriving Eq. (43) from first principles—starting from the matter stress-energy tensor in a curved background and computing the full one-loop correction to the vacuum energy density including all gravitational and gauge interactions—has not been carried out.

Conclusion. Equation (43) is a physically motivated estimate, not a derived result. Its verification or refutation through an explicit calculation—for example, computing the renormalized stress-energy tensor of a fermion field in the presence of both a curved background and a classical matter source—is identified as the most important open problem arising from this paper.

Peer Reviews and Revision History

First Round

Reviewer recommendation: Major revision required

The reviewer’s report identified four major issues and four moderate issues. Below we summarize each point and describe the changes made in response.

Major Issue 1: The central estimate $|\alpha| \sim 10^{-2}$ does not follow from the formalism presented.

The reviewer noted that channel (a) (curvature-dependent effective action) is rigorously derived and yields $|\alpha| \sim 10^{-34}$, while channel (b) (vacuum polarization) is introduced through a qualitatively different, less rigorous argument. The step connecting ρ_m as a gravitational source to ρ_m as an external source in vacuum polarization diagrams is not established with rigor.

Response and changes: We agree that this is the paper’s most important limitation. The revised manuscript now: (i) Restructures Section 6 to clearly separate channel (a) (Proposition 6.3, labeled “rigorous”) from channel (b) (now explicitly labeled “Conjecture 6.4” rather than “Proposition”). (ii) Adds a new subsection “Summary: two regimes” that states unambiguously: “The central quantitative claim of this paper is that channel (b) exists and is of the estimated magnitude, but this claim has the status of a conjecture, not a theorem.” (iii) Expands Appendix B to explicitly identify the rigorous steps, the conjectural step, and the open problem. (iv) Updates the Discussion, Limitations, and Future Directions to foreground this gap as the paper’s central open problem.

Major Issue 2: The sign ambiguity undermines the claim of a definite physical prediction.

The reviewer noted that Remark 4.4 presented $\alpha > 0$ as the physically expected case, while the RVM gives the opposite sign, creating an internal inconsistency.

Response and changes: Remark 4.4 has been rewritten to present both signs symmetrically. The language asserting $\alpha > 0$ as the “physical expectation” has been removed. The revised text states: “The sign of α is not uniquely determined by the present analysis: $\alpha > 0$ corresponds to vacuum suppression by matter, while $\alpha < 0$ corresponds to vacuum enhancement. Both possibilities are explored.” The Limitations section now explicitly acknowledges that a model unable to predict the sign of its only free parameter has limited theoretical content.

Major Issue 3: Theorem 6.2 is weaker than presented.

The reviewer requested explicit self-consistency conditions, estimation of gradient terms, and clarification of the spatial scales over which linearization is valid.

Response and changes: Theorem 6.2 has been substantially strengthened: (i) A new assumption (iv) states the self-consistency condition $|\alpha| \ll 1$ explicitly. (ii) Assumption (ii) now includes specific gradient conditions: $|\nabla \rho_m|/(m \rho_m) \ll 1$. (iii) The error terms in Eq. (20) now separate the a_2 (quadratic curvature) contributions from the gradient ($\square R$) contributions. (iv) The proof now estimates the domain of validity explicitly, noting that the linearization holds for matter varying on scales $L \gg m^{-1}$ and densities satisfying $8\pi G \rho_m \ll m^2$, while warning that halo-edge gradients require case-by-case estimation.

Major Issue 4: Theorem 7.4 (structural convergence) overstates the evidence.

The reviewer noted that the convergence is obtained only after framework-specific substitutions, and that calling this a “theorem” is inappropriate.

Response and changes: The former “Theorem 7.4” has been downgraded to Remark 7.4 (“Structural analogy across frameworks”) with accompanying argument rather than a formal proof. The text now explicitly acknowledges that the framework-specific substitutions operate on different backgrounds, that the physical content differs (vacuum energy vs. scalar dark energy), and that the analogy “does not constitute a proof of convergence” but “suggests that the matter-linear correction is a robust structural feature.”

Moderate Issue 5: Conflation of “vacuum energy” and “dark energy.”

The reviewer requested clarification of whether the vacuum correction enters as a $w = -1$ background contribution or as a matter-coupling modification.

Response and changes: A new paragraph has been added at the beginning of Section 8 (“A note on the physical nature of the correction”) that explicitly distinguishes the two cases, notes that the present analysis does not fully resolve this distinction, and identifies the specification of the perturbed stress-energy tensor as a necessary next step.

Moderate Issue 6: Observational section too preliminary.

The reviewer noted that the growth-rate estimate needs to be verified by solving the growth equation, and that the distinction between a $w = -1$ modification and a matter-coupling rescaling produces different signatures.

Response and changes: The perturbation-theory subsection now explicitly states that the growth-rate estimate “has not been verified by numerically solving the growth equation” and that “the actual magnitude depends on whether the vacuum correction clusters or remains homogeneous.” The Section Result has been updated accordingly.

Moderate Issue 7: Missing discussion of the Equivalence Principle.

Response and changes: A new counterargument-and-response pair has been added to the Discussion section addressing the strong equivalence principle. The response uses the Casimir effect as a precedent, notes the subtlety of spatial vacuum-energy gradients, and defers a detailed SEP analysis (including lunar laser ranging and pulsar timing tests) to future work.

Moderate Issue 8: Appendix B is incomplete.

Response and changes: Appendix B has been completely rewritten. It now identifies three steps (Step 1 rigorous, Step 2 rigorous in flat space, Step 3 conjectural), explicitly derives the factor $1/(48\pi^2)$ from $\Pi(0)$, and states clearly that the final equation is “a physically motivated estimate, not a derived result.”

Minor issues.

Eq. (16) drops the vacuum contribution: The condition $\rho_{\text{vac}} \ll \rho_{\text{m}}$ is now stated explicitly, with a note that it is satisfied in matter-dominated environments but not at the present-day cosmic mean.

Self-contained references: A new limitation (seventh) notes that program references provide context but are not logically required.

Brown reference status: Ref. [26] is confirmed published: Int. J. Mod. Phys. D **35**(4), 2650006 (2026), DOI: 10.1142/S0218271826500069, Open Access.

Peer review placeholders: Removed; this section now documents the actual review.